

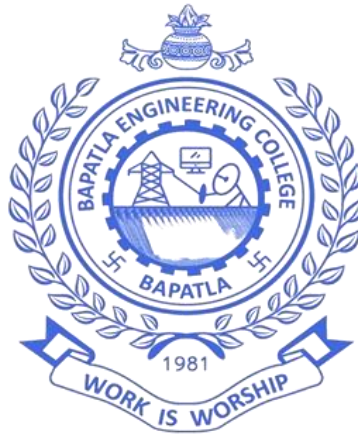
A Project Report on
Smart Travelling Assistant
Submitted in partial fulfillment for award of
Bachelor of Technology
in
Computer Science and Engineering

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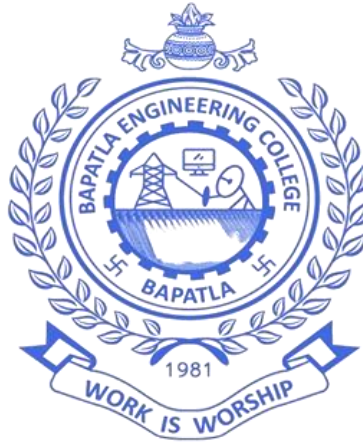
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CERTIFICATE

This is to certify that the project report entitled **Smart Travelling Assistant** that is being submitted by K.Asha (Y21ACS476), G.M.N.Bramham (Y22ACS453), P.Dinesh (L23ACS599), G.Naga Ganesh (L23ACS602) in partial fulfillment for the award of the Degree of Bachelor of Technology in Computer Science and Engineering to the Acharya Nagarjuna University is a record of bonafide work carried out by them under our guidance and supervision.

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DECLARATION

We declare that this project work is composed by ourselves, that the work contained herein is our own except where explicitly stated otherwise in the text, and that this work has not been submitted for any other degree or professional qualification except as specified.

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TABLE OF CONTENTS

LIST OF TABLES.....	vii
LIST OF FIGURES.....	viii
LIST OF ABBREVIATIONS.....	ix
ABSTRACT.....	x
1. INTRODUCTION.....	1
1.1 MOTIVATION	1
1.2 PROBLEM STATEMENT	2
1.3 OBJECTIVE OF PROJECT	3
2. LITERATURE SURVEY	4
2.1 EXISTING SYSTEM	5
2.2 DISADVANTAGES OF EXISTING SYSTEM	6
2.3 PROPOSED SYSTEM.....	6
2.4 ADVANTAGESOF PROPOSED SYSTEM.....	7
3. ANALYSIS	9
3.1 SOFTWARE REQUIREMENT SPECIFICATION	9
3.1.1 User Requirements.....	9
3.1.2 Software Requirements	10
3.1.3 Hardware Requirements	11
3. 2 CONTENT DIAGRAM OF PROJECT	12
4. DESIGN	16
4.1 INTRODUCTION.....	16
4.2 DFD / ER / UML DIAGRAM	16
4.3 DATA DICTIONARY	23
5. IMPLEMENTATION & RESULTS	25
5.1 METHOD OF IMPLEMENTATION	25
5.1.1 Implementation of Algorithms.....	25
5.1.2 Output Screens	26
5.1.3 Result Analysis	32
6. TESTING & VALIDATION	35
6.1 FEATURES TO BE TESTED	35

6.2 FEATURES NOT TO BE TESTED	36
6.3 TESTING TOOLS AND ENVIRONMENT	36
6.4 DESIGN OF TEST CASES AND SCENARIOS	37
7. CONCLUSION& FUTURE ENHANCEMENT	39
7.1 PROJECT CONCLUSION	39
7.2 FUTURE ENHANCEMENT	39
8. BIBLIOGRAPHY	41

LIST OF TABLES

Table 4.1 User Collection	23
Table 4.2 OTP Collection	24
Table 4.3 TripPlan Collection.....	24
Table 6.1 Test Cases.....	37

LIST OF FIGURES

Figure 3.1 Content Diagram.....	12
Figure 4.1 Use Case Diagram.....	16
Figure 4.2 Sequence Diagram.....	18
Figure 4.3 Class Diagram.....	20
Figure 4.4 Data Flow Diagram.....	21
Figure 5.1 Landing Page.....	26
Figure 5.2 Login Page.....	27
Figure 5.3 Register Page.....	27
Figure 5.4 Input From Page.....	28
Figure 5.5 Transportation Page Part 1.....	28
Figure 5.6 Transportation Page Part 2.....	29
Figure 5.7 Places Page Part 1.....	29
Figure 5.8 Places Page Part 2.....	30
Figure 5.9 Accomodation Page.....	30
Figure 5.10 Budget Page.....	31
Figure 5.11 Profile page.....	31
Figure 5.12 Itinerary Page Part 1.....	33
Figure 5.13 Itinerary Page Part 2.....	34
Figure 5.14 Itinerary Page Part 3.....	34

LIST OF ABBREVIATIONS

API – Application Programming Interface

OTP – One Time Password

REST – Representational State Transfer

KNN – K-Nearest Neighbors

VGG-16 – Visual Geometry Group 16-layer Network

ML – Machine Learning

SRS – Software Requirement Specification

HTML – HyperText Markup Language

CSS – Cascading Style Sheets

UI – User Interface

CORS – Cross-Origin Resource Sharing

NoSQL – Not Only SQL

RAM – Random Access Memory

JWT – JSON Web Token

CSV – Comma-Separated Values

JSON – JavaScript Object Notation

ORS – OpenRouteService

UML – Unified Modeling Language

DFD – Data Flow Diagram

ER – Entity Relationship

TTL – Time To Live

ABSTRACT

This project focuses on the development of a user-friendly web-based application, **Smart Travelling Assistant**, designed to simplify and enhance the travel planning process. The system automatically generates personalized travel itineraries based on user inputs such as destination, budget, travel dates, and number of travelers, thereby reducing manual effort and improving planning efficiency.

The application is developed using a modern full-stack architecture, with React for building a dynamic and responsive frontend interface, and Node.js with Express for handling backend operations and RESTful services. MongoDB Atlas is utilized as a cloud-based database to securely manage user and travel data. The system incorporates robust security features, including Google OAuth authentication, OTP-based verification, and Bcrypt-based password encryption, ensuring data privacy and protection.

To provide accurate and real-time travel information, the system integrates multiple external APIs such as AviationStack for flight data, Geoapify for location services, OpenRouteService for route optimization, and Google Places for accommodation and attraction details. Additionally, middleware technologies like CORS and Multer are employed to support secure communication and efficient data handling.

A key feature of the system is its ability to generate structured, day-by-day travel itineraries using intelligent planning logic, ensuring practical and time-efficient schedules. The application also includes a user dashboard for profile management and travel history tracking. Deployed using modern cloud platforms, the system ensures scalability, accessibility, and reliability.

Overall, the Smart Travelling Assistant provides a comprehensive and integrated solution for travel planning by combining real-time data, intelligent processing, and user-centric design, thereby significantly improving the overall travel experience.

1. INTRODUCTION

Travel has become an essential part of modern life for purposes such as tourism, business, education, and personal activities. Planning a trip involves several tasks including selecting destinations, arranging transportation, accommodations, managing budgets, and organizing schedules. Performing these activities manually can be time-consuming and often leads to inefficiencies, making the process complex for travelers.

Traditional travel planning methods require users to gather information from multiple sources such as websites and applications. This scattered approach can result in incomplete or unorganized plans, increasing the chances of errors and inconvenience. Therefore, there is a need for a unified system that can simplify and streamline the entire travel planning process.

With advancements in web technologies and cloud-based systems, automated travel planning solutions have become more efficient and accessible. These systems use modern frameworks and APIs to collect, process, and present travel-related information in a structured and user-friendly manner. They help users make better decisions by providing accurate and real-time data.

The **Smart Travelling Assistant** is developed as a web-based application that automates itinerary generation based on user preferences such as destination, travel dates, budget, and number of travelers. It integrates a React-based frontend with a Node.js and Express backend, along with MongoDB Atlas for secure data storage and management.

Furthermore, the system incorporates external APIs to provide real-time information on routes, locations, flights, and accommodations. It also includes secure authentication features such as Google OAuth and OTP verification. Overall, the application simplifies travel planning, improves efficiency, and enhances the overall user experience.

1.1 MOTIVATION

Travel planning often involves handling multiple tasks such as searching for destinations, comparing travel options, managing budgets, and organizing schedules.

This process can be time-consuming and confusing, especially when information is scattered across different platforms. The difficulty in managing all these aspects efficiently motivated the need for a unified system that simplifies travel planning.

Another key motivation behind this project is the increasing demand for personalized and user-friendly digital solutions. Modern users expect quick access to relevant information and customized recommendations based on their preferences. Existing systems often lack proper integration of services, leading to inconvenience and reduced efficiency. This inspired the development of a solution that combines multiple travel services into a single platform.

Additionally, advancements in web technologies and API integrations provide an opportunity to build intelligent systems that can automate complex tasks. By leveraging these technologies, it is possible to create a system that not only saves time but also improves decision-making and user experience.

Overall, the motivation of this project is to develop an efficient, reliable, and easy-to-use travel assistant that reduces manual effort, enhances convenience, and provides a seamless travel planning experience for users.

1.2 PROBLEM STATEMENT

Travel planning is a complex and time-consuming process that requires users to gather information from multiple sources such as websites and applications for destinations, transportation, accommodation, and budgeting. This fragmented approach often leads to inefficiencies, increased effort, and unorganized travel plans. Users may face difficulties in comparing options, managing schedules, and accessing accurate real-time information.

Existing travel platforms typically focus on specific services, such as booking or location search, rather than providing a complete and integrated solution. As a result, users are required to switch between multiple platforms, which increases complexity and reduces overall efficiency. Additionally, many systems lack proper personalization, making it difficult for users to obtain travel plans tailored to their preferences.

There is a need for a unified system that can automate and simplify the travel planning process by integrating multiple services into a single platform. Such a system should provide personalized itineraries, real-time travel information, secure user authentication, and an intuitive interface. The proposed Smart Travelling Assistant aims to address these challenges by offering an efficient, reliable, and user-friendly solution for modern travel planning.

1.3 OBJECTIVE OF PROJECT

The main objective of the **Smart Travelling Assistant** is to simplify and automate the travel planning process by providing users with personalized and efficient trip plans. The system aims to reduce the complexity of organizing travel by integrating multiple services into a single platform, ensuring convenience, accuracy, and a better user experience.

- To develop a user-friendly and responsive web application for travel planning
- To generate personalized travel itineraries based on user preferences
- To integrate multiple external APIs for real-time travel data (routes, flights, locations, accommodation)
- To implement secure authentication using Google OAuth and OTP verification
- To store and manage user and travel data efficiently using MongoDB Atlas
- To provide a dashboard for users to manage profiles and view travel history
- To ensure smooth communication between frontend and backend using REST APIs
- To design a scalable system that supports future enhancements like AI recommendations and booking features

2. LITERATURE SURVEY

In recent years, significant research has been carried out in the field of intelligent travel recommendation systems to improve user experience and automate trip planning. According to the International Research Journal of Modernization in Engineering Technology and Science (IRJMETS), 2022, the rapid growth of the tourism industry has increased the demand for smart systems that can assist users in selecting destinations and organizing travel efficiently. The study titled “Smart Travel Assistant with Itinerary Planner using Hybrid Machine Learning Approach” by **S. Sangeetha Mariammal et al. (2022)** highlights the challenges faced by users in handling large volumes of travel data and emphasizes the need for intelligent recommendation systems.

The authors propose a hybrid machine learning-based system that integrates multiple algorithms such as K-Means Clustering, K-Nearest Neighbors (KNN), and Logistic Regression to generate personalized travel recommendations. K-Means clustering is used to group nearby tourist attractions, while KNN is applied to map suitable hotels based on proximity. Logistic Regression is utilized for sentiment analysis of user reviews, enabling the system to refine recommendations based on feedback. Additionally, transfer learning using the VGG-16 model is employed for image-based location identification, enhancing the system’s functionality.

The research further discusses the importance of data collection from reliable sources such as TripAdvisor, Holidify, and Yatra, where web scraping techniques are used to gather information about tourist attractions, hotels, ratings, and reviews. A derived metric called “total score” is calculated using ratings and the number of reviews to evaluate the importance of each location. This structured dataset is then used to train machine learning models and generate meaningful travel itineraries.

Another important contribution of the study is the implementation of route optimization techniques using the 2-Opt algorithm, which helps in minimizing travel distance and improving itinerary efficiency. The system architecture, illustrated in Figure 1 on page 2, shows a modular design consisting of components such as place recommendation, user preference module, itinerary generation, image finder, and

feedback analysis. These modules work together to provide a comprehensive travel planning solution.

The study also highlights the role of user interaction in improving recommendation accuracy. The feedback and review module uses sentiment analysis to understand user satisfaction and dynamically adjust future recommendations. Positive feedback results in suggesting similar destinations, while negative feedback helps refine the system. This adaptive approach enhances personalization and makes the system more user-centric.

In addition to this work, several researchers have explored recommendation systems based on sentiment analysis, route optimization, and image classification. For instance, **Achin Jain et al. (2016)** studied recommendation systems using sentiment analysis, emphasizing the role of user feedback in improving system accuracy. Similarly, **Catherine Almira et al. (2019)** proposed travel itinerary recommendation using optimization techniques, and **Srikanth Tammina (2019)** demonstrated the effectiveness of transfer learning models like VGG-16 in image classification tasks. These studies collectively highlight the importance of combining machine learning techniques for intelligent system design.

Despite these advancements, existing systems often face challenges such as high computational complexity, dependency on large datasets, and limited real-time usability. Many solutions are research-oriented and lack simplicity in design and implementation. The proposed **Smart Travelling Assistant** builds upon these research contributions while focusing on a simpler, scalable, and user-friendly approach using modern web technologies and API integrations. This ensures better practicality, improved performance, and enhanced user experience.

2.1 EXISTING SYSTEM

The existing travel planning systems primarily rely on traditional methods or partially automated platforms that provide limited services such as destination search, hotel booking, or route information. Most of these systems require users to manually gather information from multiple sources, including websites and mobile applications,

to plan their trips. This fragmented approach makes the process time-consuming and less efficient.

Some advanced systems use machine learning techniques to provide recommendations and itinerary planning. These systems analyze user preferences, reviews, and travel data to generate suggestions. However, they often involve complex algorithms and require large datasets for training, which increases system complexity and limits real-time usability.

Additionally, many existing systems lack proper integration of services, forcing users to switch between different platforms for transportation, accommodation, and itinerary management. This reduces user convenience and makes travel planning more complicated than necessary.

2.2 DISADVANTAGES OF EXISTING SYSTEM

- Requires users to rely on multiple platforms for complete travel planning
- Time-consuming and inefficient due to manual data collection
- Limited personalization in many systems
- High system complexity in ML-based solutions
- Requires large datasets and computational resources
- Not fully suitable for real-time applications
- Lack of user-friendly interface and seamless integration
- Increased chances of errors and incomplete planning

2.3 PROPOSED SYSTEM

The proposed system, **Smart Travelling Assistant**, is designed to overcome the limitations of existing travel planning methods by providing a unified, efficient, and user-friendly platform. It is a web-based application that automates the process of travel planning by generating personalized itineraries based on user inputs such as destination, budget, travel dates, and number of travelers.

The system integrates a modern frontend developed using React with a robust backend built using Node.js and Express. MongoDB Atlas is used as a cloud-based

database to securely store user information and travel data. The application follows a RESTful architecture to ensure smooth communication between the frontend and backend components.

To enhance functionality, the system utilizes various external APIs to provide real-time travel information, including route optimization, location search, flight details, and accommodation suggestions.

This enables users to access accurate and up-to-date information within a single platform, eliminating the need to switch between multiple applications.

The proposed system also includes secure authentication mechanisms such as Google OAuth and OTP-based verification to ensure user data protection. Additional features such as user profile management, travel history tracking, and a responsive user interface further improve usability and overall user experience.

Overall, the Smart Travelling Assistant focuses on simplicity, efficiency, and scalability. It reduces manual effort, improves decision-making, and provides a seamless travel planning experience. The system is also designed to support future enhancements such as AI-based recommendations, booking services, and payment integration.

2.4 ADVANTAGES OF PROPOSED SYSTEM

The proposed **Smart Travelling Assistant** provides a comprehensive and efficient solution for modern travel planning by addressing the limitations of existing systems. It offers an integrated platform that enhances usability, performance, and user satisfaction through the following advantages:

- **Unified and Integrated Platform:**

The system consolidates multiple travel-related services such as itinerary planning, route optimization, and accommodation suggestions into a single application, eliminating the need for users to rely on multiple platforms.

- **Enhanced Time Efficiency:**

By automating the travel planning process, the system significantly reduces the time and effort required to organize trips, enabling users to generate complete travel plans instantly.

- **Personalized User Experience:**

The application delivers customized travel itineraries based on individual user preferences, including destination, budget, and travel duration, thereby improving relevance and user satisfaction.

- **Real-Time Data Availability:**

Integration with external APIs ensures that users receive accurate, dynamic, and up-to-date information related to travel routes, locations, and services.

- **User-Centric Interface Design:**

The system is developed with a simple, intuitive, and responsive interface, making it accessible to users with varying levels of technical expertise.

- **Reduction in Manual Effort:**

The automation of data collection and itinerary generation minimizes manual intervention, reducing errors and improving overall efficiency.

- **Scalability and Extensibility:**

The system architecture supports future enhancements such as AI-driven recommendations, booking integrations, and mobile platform expansion.

- **Improved Decision Support:**

By presenting structured and well-organized travel plans, the system assists users in making informed and effective travel decisions

3. ANALYSIS

3.1 SOFTWARE REQUIREMENT SPECIFICATION

The Software Requirement Specification (SRS) provides a detailed description of the functional and non-functional requirements of the **Smart Travelling Assistant**. It serves as a foundation for system design and development by clearly defining system behavior, user expectations, and operational constraints.

The SRS ensures that all stakeholders, including developers and users, have a clear understanding of the system's objectives and functionalities.

3.1.1 User Requirements

User requirements define the expectations and needs of end-users interacting with the system. The Smart Travelling Assistant is designed to provide a seamless and efficient travel planning experience through the following user requirements:

- The system should allow users to **register and login securely** using email/password and Google OAuth authentication.
- Users must be able to **input travel details**, including destination, travel dates, budget, and number of travelers.
- The system should generate **personalized and structured day-wise travel itineraries**, ensuring practical and optimized schedules.
- Users should be able to **view recommended accommodations**, including hotel details such as ratings, location, and estimated cost.
- The system must provide **route details and travel paths**, including distance and estimated travel time between locations.
- A **user dashboard** should be available to manage profile information, preferences, and travel history.
- The system should allow users to **save, view, and delete previous travel plans** for future reference.
- The application interface should be **user-friendly, responsive, and easy to navigate**, ensuring accessibility for all types of users.
- The system should provide **real-time information** to ensure accuracy and reliability of travel plans.

3.1.2 Software Requirements

Software requirements define the technical stack, tools, and frameworks used to develop and run the system efficiently.

Frontend Requirements

- React.js for building dynamic and responsive user interfaces
- HTML, CSS, and JavaScript for UI design and styling
- Axios or Fetch API for handling API requests

Backend Requirements

- Node.js for server-side execution
- Express.js framework for building RESTful APIs
- Middleware such as CORS for secure communication and Multer for file handling

Database Requirements

- MongoDB Atlas (cloud-based NoSQL database)
- Used for storing user credentials, travel plans, and history data

API Integrations

- AviationStack API → Provides real-time flight data
- Geoapify API → Location search and attraction data
- OpenRouteService API → Route optimization and distance calculation
- Google Places API → Hotel and accommodation details

Authentication & Security

- Google OAuth 2.0 for secure login
- OTP verification using SendGrid or similar services
- Bcrypt for password encryption
- Secure API key management using environment variables

Deployment Requirements

- Frontend deployed on **Vercel**
- Backend deployed on **Render**
- Internet connectivity required for accessing APIs and services

System Compatibility

- Compatible with major web browsers (Chrome, Firefox, Edge)
- Cross-platform support (Windows, Linux, macOS)

3.1.3 Hardware Requirements

The hardware requirements for the system are minimal, making it accessible and cost-effective.

Development Environment

- Processor: Intel i3 or higher
- RAM: Minimum 4 GB (8 GB recommended for better performance)
- Storage: At least 20 GB of free disk space
- Stable internet connection for API integration and deployment

User Environment

- Device: Smartphone, Laptop, or Desktop
- Web browser with internet connectivity
- Basic system configuration sufficient for running web applications

3.2 CONTENT DIAGRAM OF PROJECT

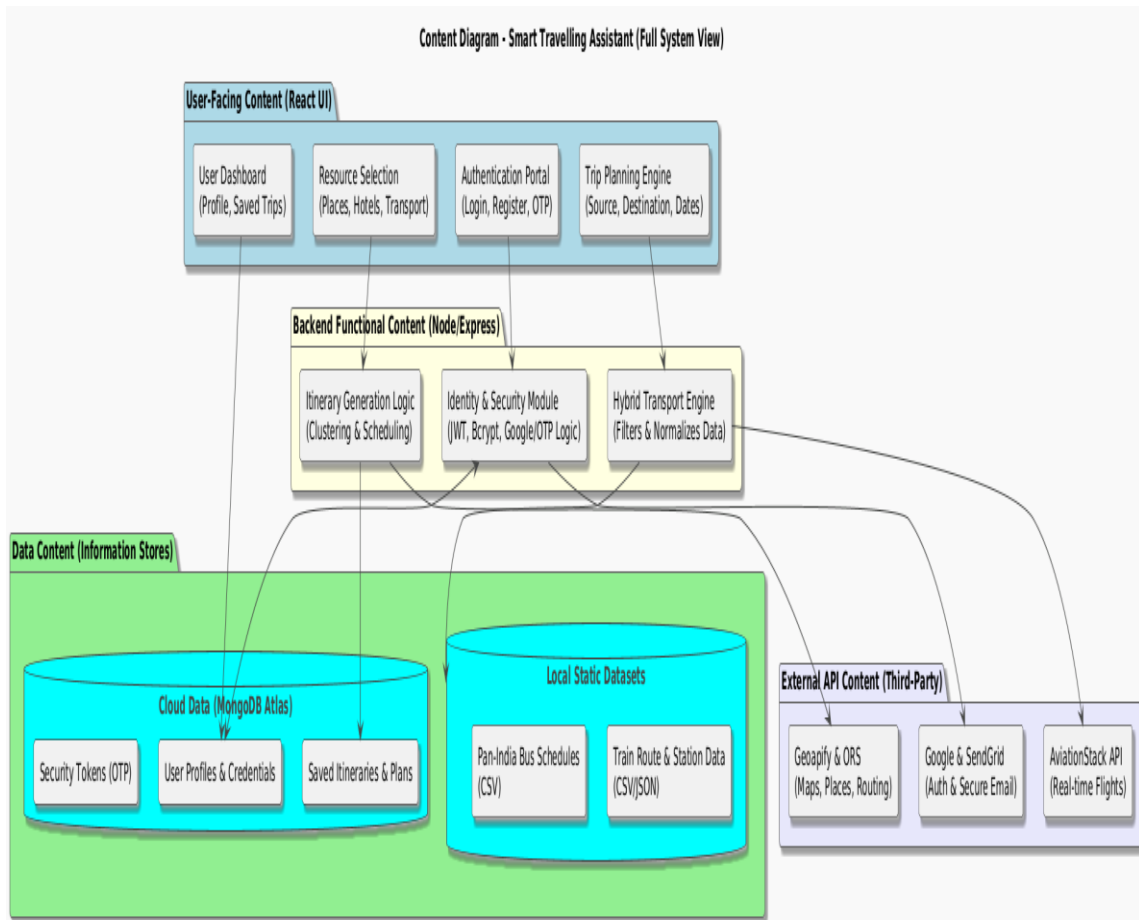


Figure 3.1 Content Diagram

Overview

The Content Diagram represents the overall architecture of the Smart Traveling Assistant system, showing how different components interact. It is divided into four major layers:

- User-Facing Content (Frontend)
- Backend Functional Content
- Data Content (Storage)
- External API Content (Third-Party Services)

This layered structure ensures modularity, scalability, and efficient data flow.

User-Facing Content (React UI)

This layer represents the frontend interface developed using React. It allows users to interact with the system.

Components:

- **User Dashboard (Profile, Saved Trips):**
Displays user information, previous trips, and saved itineraries.
- **Resource Selection (Places, Hotels, Transport):**
Enables users to choose destinations, accommodations, and transportation options.
- **Authentication Portal (Login, Register, OTP):**
Handles user authentication with secure login, registration, and OTP verification.
- **Trip Planning Engine (Source, Destination, Dates):**
Accepts user inputs and initiates trip planning.

Backend Functional Content (Node.js / Express)

This layer processes user requests and applies business logic.

Modules:

- **Itinerary Generation Logic (Clustering & Scheduling):**
Generates optimized travel plans using algorithms for route clustering and scheduling.
- **Identity & Security Module (JWT, Bcrypt, Google/OTP Logic):**
Ensures secure authentication and data protection using token-based authentication and encryption.
- **Hybrid Transport Engine (Filters & Normalizes Data):**
Processes and filters transport data from multiple sources to provide the best options.

Data Content (Information Stores)

This layer stores both dynamic and static data required for the application.

Cloud Data (MongoDB Atlas):

- Security Tokens (OTP)
- User Profiles & Credentials
- Saved Itineraries & Plans

Local Static Datasets:

- Pan-India Bus Schedules (CSV)
- Train Routes & Station Data (CSV/JSON)

These datasets help in offline support and faster processing.

External API Content (Third-Party Services)

This layer integrates third-party services to enhance functionality.

APIs Used:

- Geoapify & ORS (Maps, Places, Routing):
Provides geolocation, route optimization, and place suggestions.
- Google & SendGrid (Auth & Secure Email):
Handles authentication and email notifications.
- AviationStack API (Real-time Flights):
Supplies real-time flight data.

Data Flow Description

- User interacts with the React UI.
- Requests are sent to the backend server.
- Backend processes data using business logic modules.
- Data is fetched/stored in MongoDB or local datasets.
- External APIs are called when required.
- Processed results are returned to the user interface.

Key Features Highlighted

- Modular architecture
- Secure authentication system
- Real-time and static data integration
- Intelligent trip planning
- Scalable backend design

4. DESIGN

4.1 INTRODUCTION

The design phase is a crucial stage in the software development lifecycle, where the system requirements identified during the analysis phase are translated into a structured and organized solution. It provides a blueprint for the implementation of the system by defining its architecture, components, data flow, and interactions.

The **Smart Travelling Assistant** is designed using a modular and scalable approach to ensure flexibility, maintainability, and efficient performance. The design focuses on integrating frontend, backend, database, and external APIs in a seamless manner to deliver a smooth and user-friendly travel planning experience.

This chapter presents various design aspects of the system, including Data Flow Diagrams (DFD), Unified Modeling Language (UML) diagrams, and the Data Dictionary, which collectively provide a comprehensive understanding of the system structure and functionality.

4.2 DFD / ER / UML DIAGRAM

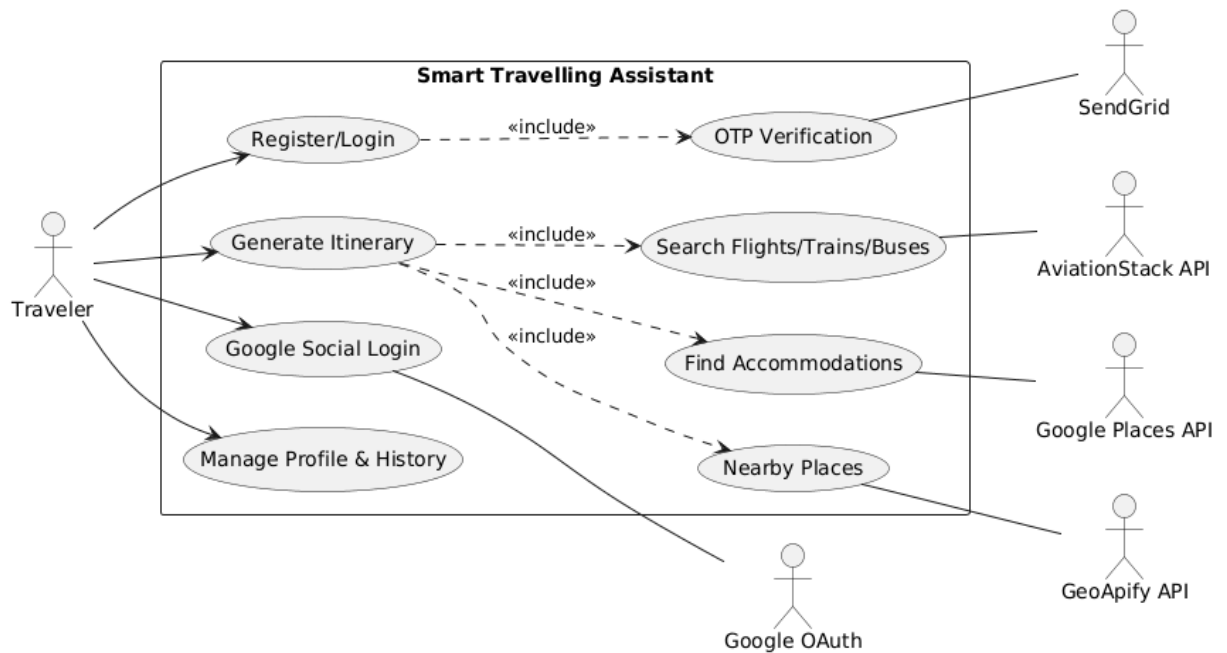


Figure 4.1 Use Case Diagram

Overview

The Use Case Diagram illustrates the interaction between the traveler and the Smart Travelling Assistant system, highlighting the key functionalities involved in travel planning and management.

Use Cases

- **User Registration and Login**

Enables users to create an account and securely access the system.

- **OTP Verification**

Provides secure authentication through one-time password validation.

- **Google Authentication**

Allows users to log in using their Google account.

- **Itinerary Generation**

Generates a complete travel plan based on user inputs such as destination and dates.

- **Transport Search**

Retrieves available options for flights, trains, and buses.

- **Accommodation Search**

Suggests suitable hotels and lodging options.

- **Nearby Places Recommendation**

Displays nearby attractions and points of interest.

Relationships

<<include>> Relationship

- Login includes OTP Verification
- Itinerary Generation includes Transport Search, Accommodation Search, and Nearby Places Recommendation

Summary

The Use Case Diagram presents a structured view of system functionalities, emphasizing secure user authentication, integrated travel services, and efficient trip planning within a unified platform.

Sequence Diagram – Authentication Process

Overview

The sequence diagram shows the interaction between user, frontend, backend, database, and external services during authentication.

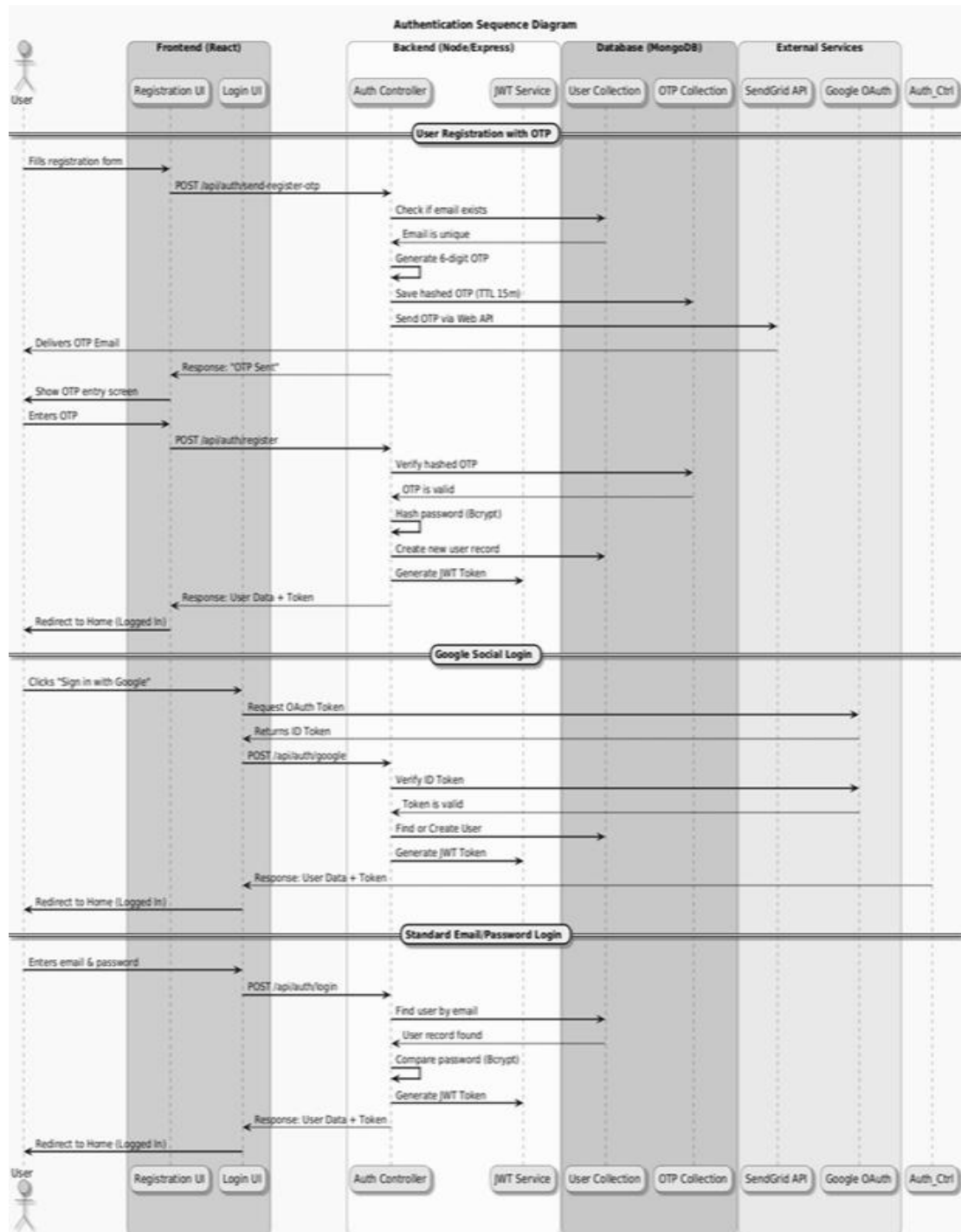


Figure 4.2 Sequence Diagram

User Registration with OTP

- User submits registration form
- Backend checks email and generates OTP
- OTP is stored (hashed) and sent via email
- User enters OTP for verification
- Password is encrypted and user is created
- JWT token is generated and user is logged in

Google Social Login

- User selects “Sign in with Google”.
- Frontend requests authentication from Google OAuth.
- Google returns an ID token.
- Frontend sends the token to the backend.
- Backend verifies the token using Google services.
- If valid, the system checks for an existing user or creates a new user.
- JWT token is generated.
- User is logged in and redirected to the home page.

Standard Login

- User enters email and password.
- Frontend sends login request to the backend.
- Backend retrieves user data from the database.
- Password is verified using Bcrypt comparison.
- If credentials are valid, a JWT token is generated.
- User is logged in and redirected to the home page.

Key Features

- Secure authentication using OTP and JWT
- Password encryption using Bcrypt
- Integration with external services (SendGrid, Google OAuth)
- Multiple login options for user convenience
- Efficient session handling with token-based authentication

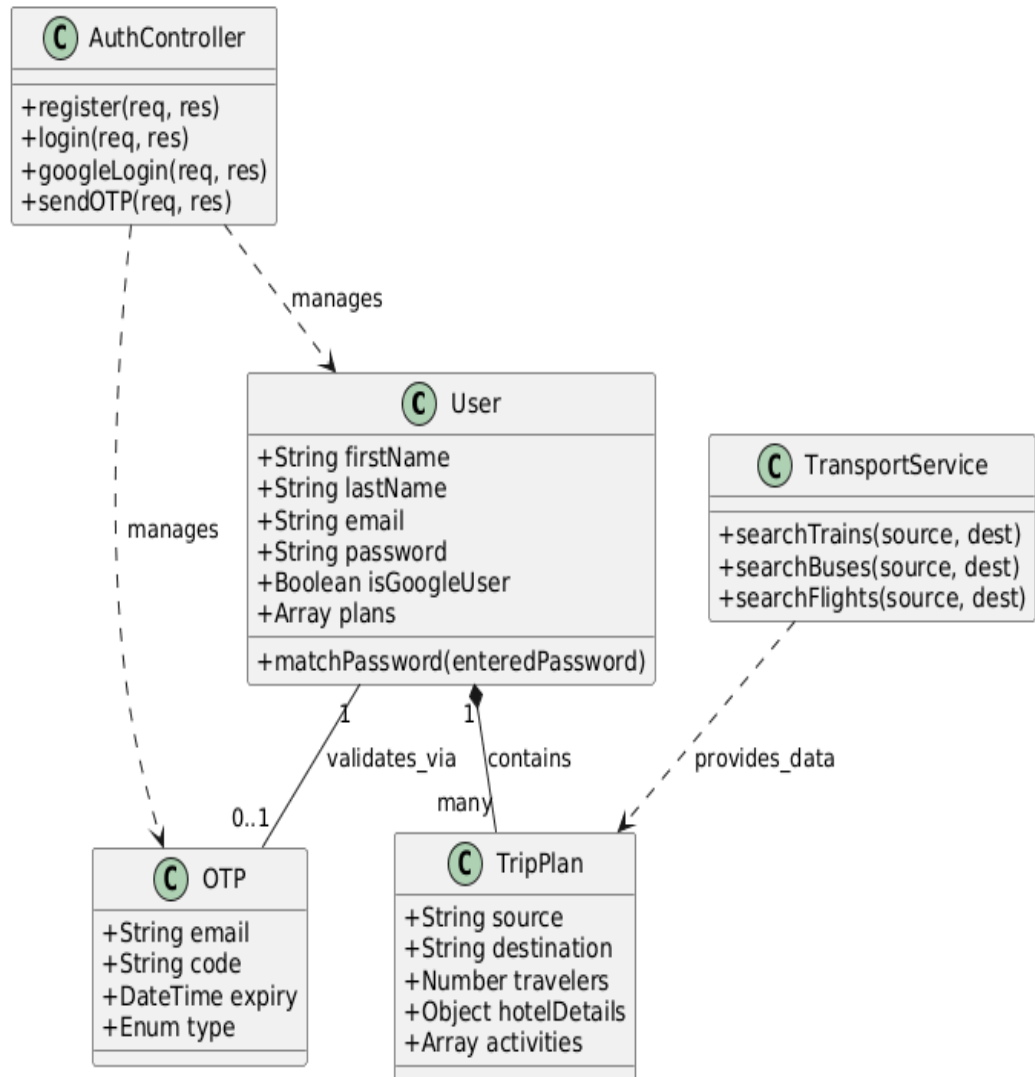


Figure 4.3 Class Diagram

Overview

The class diagram represents the static structure of the system, showing the main classes, their attributes, methods, and relationships involved in authentication and trip planning.

Classes

- **AuthController:**
Manages registration, login, Google login, and OTP handling.
- **User:**
Stores user details such as name, email, password, and trip plans; includes password validation.

- **OTP:**
Handles OTP data for user verification with expiry.
- **TripPlan:**
Represents travel plans including source, destination, travelers, and activities.
- **TransportService:**
Provides methods to search trains, buses, and flights.

Relationships (Elaborated)

- AuthController manages User and OTP
- User is validated using OTP (0..1)
- User can have multiple TripPlans (1..*)
- TransportService provides data to TripPlan

Summary

The diagram presents a clear structure of system components, focusing on user authentication and travel planning with well-defined relationships.

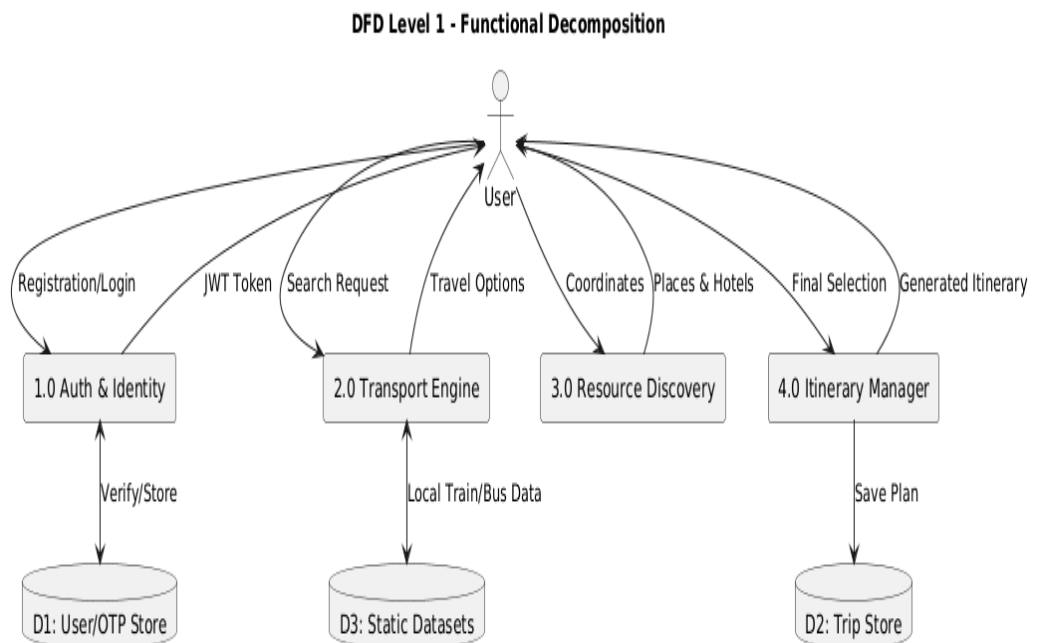


Figure 4.4 Data Flow Diagram

Overview

The Level 1 Data Flow Diagram (DFD) represents the detailed functional decomposition of the Smart Travelling Assistant system. It shows how user inputs are processed through different modules and how data flows between processes and data stores.

External Entity

- **User:**

The primary entity who interacts with the system by providing inputs and receiving travel plans.

Processes

Auth & Identity:

Handles user registration and login, verifies credentials, and generates JWT tokens for secure access.

Transport Engine:

Processes search requests and retrieves available transport options such as trains and buses using local datasets.

Resource Discovery:

Identifies places, hotels, and travel-related resources based on user inputs like coordinates and preferences.

Itinerary Manager:

Combines selected options to generate a complete travel itinerary and manages final plan creation.

Data Stores

D1: User/OTP Store:

Stores user credentials and OTP data for authentication.

D2: Trip Store:

Stores generated itineraries and user travel plans.

D3: Static Datasets:

Contains local data such as train and bus schedules.

Data Flow

- User provides registration/login details → Auth & Identity
- Auth process verifies and stores data in User/OTP Store
- User sends search request → Transport Engine
- Transport Engine fetches data from Static Datasets
- User inputs coordinates → Resource Discovery
- Resource Discovery returns places and hotel options
- User makes final selection → Itinerary Manager
- Itinerary Manager generates and saves itinerary in Trip Store
- Final itinerary is returned to the user

Summary

The DFD Level 1 diagram illustrates how the system processes user inputs through authentication, transport search, resource discovery, and itinerary generation modules. It shows the flow of data between processes and data stores, resulting in the creation and storage of a complete travel plan for the user.

4.3 DATA DICTIONARY

Table 4.1 User Collection

Attribute	Data Type	Description	Constraints
_id	ObjectId	Primary Key for the user	System Generated
Email	String	User's unique identifier	Unique, Required
Password	String	Hashed password string	Required for non-Google
Plans	Array	References to saved itineraries	Foreign Key

Table 4.2 OTP Collection

Attribute	Data Type	Description	Constraints
Email	String	Recipient of the security code	Required
Code	String	Hashed 6-digit OTP	Required
Expiry	DateTime	Automatic deletion timestamp	TTL Index (15 mins)

Table 4.3 TripPlan Collection

Attribute	Data Type	Description	Constraints
Source	String	Origin city of the trip	Required
Destination	String	Target city of the trip	Required
Activities	Array	Structured day-wise tasks	JSON Object
TotalBudget	Number	Total estimated trip cost	Calculated

5. IMPLEMENTATION & RESULTS

5.1 METHOD OF IMPLEMENTATION

The implementation of the Smart Travelling Assistant is carried out using a full-stack development approach, integrating frontend, backend, database, and external APIs. The system follows a modular architecture, ensuring scalability, maintainability, and efficient performance.

The frontend is developed using React.js, which provides a dynamic and responsive user interface. The backend is implemented using Node.js and Express, responsible for handling business logic, API communication, and request processing. The application uses MongoDB Atlas as a cloud-based database for storing user data, travel plans, and history.

The system operates in a step-by-step process:

- Users register or log in to the application
- Travel details such as destination, budget, and duration are entered
- Backend processes the input and interacts with external APIs
- Itinerary generation logic organizes travel plans
- Results are displayed in a structured format

This approach ensures smooth interaction between system components and efficient execution of functionalities.

5.1.1 Implementation of Algorithms

The core functionality of the system is based on an intelligent itinerary generation mechanism.

Time-Proximity Clustering Algorithm

The system uses a Time-Proximity Clustering Logic to generate realistic and optimized travel plans. This algorithm works as follows:

- Collects location data (latitude and longitude) from APIs
- Groups nearby attractions based on geographical proximity
- Calculates travel time and distance between locations

- Allocates places into different days based on time constraints
- Divides each day into time slots (morning, afternoon, evening)
- Ensures minimal travel time and maximum coverage of attractions

This algorithm helps in generating practical and efficient itineraries, unlike traditional systems that provide only generic recommendations.

5.1.2 Output Screens

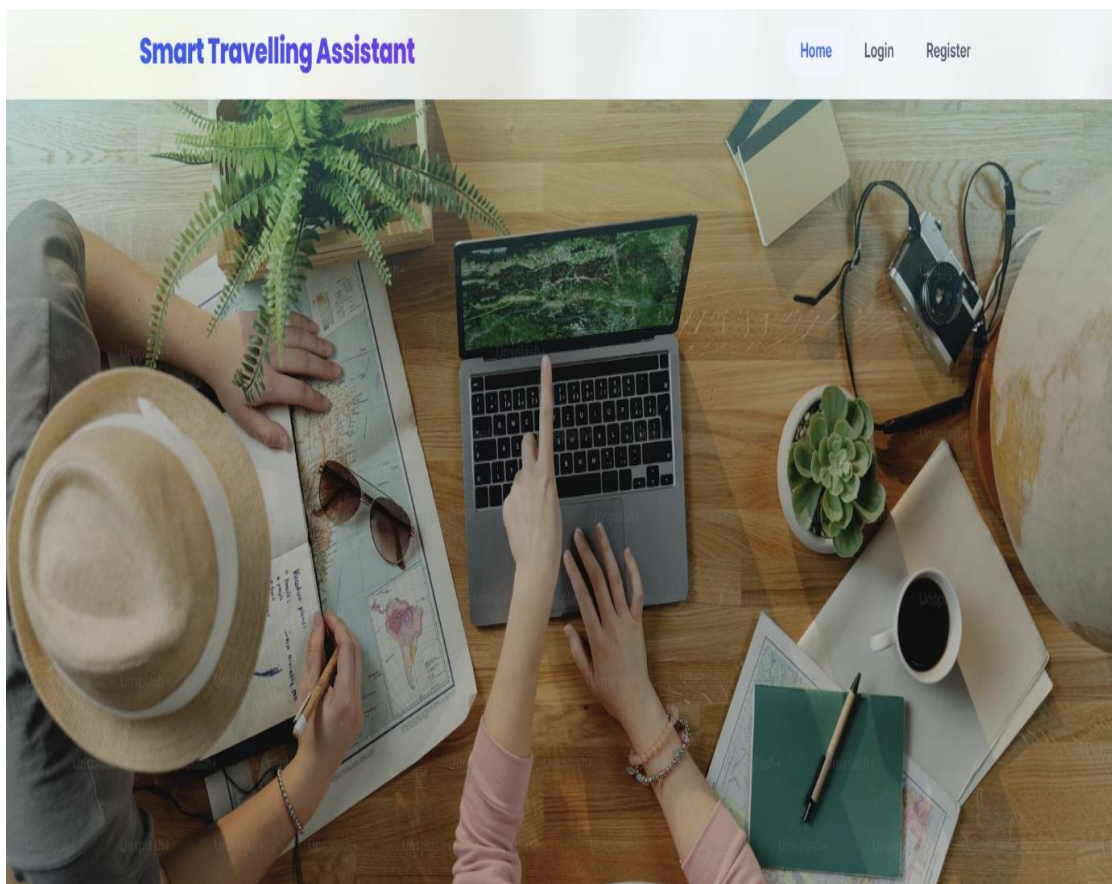


Figure 5.1 Landing Page

Smart Travelling Assistant

Login to TripPlanner

Email

Password

Login

Sign in as Mohan
mohangantu55@gmail.com

Don't have an account? [Register here](#)

Forgot your password? [Reset here](#)

Figure 5.2 Login Page

Smart Travelling Assistant

Create Account

First Name

Last Name

Username

Mobile Number

Address

Email

Date of Birth

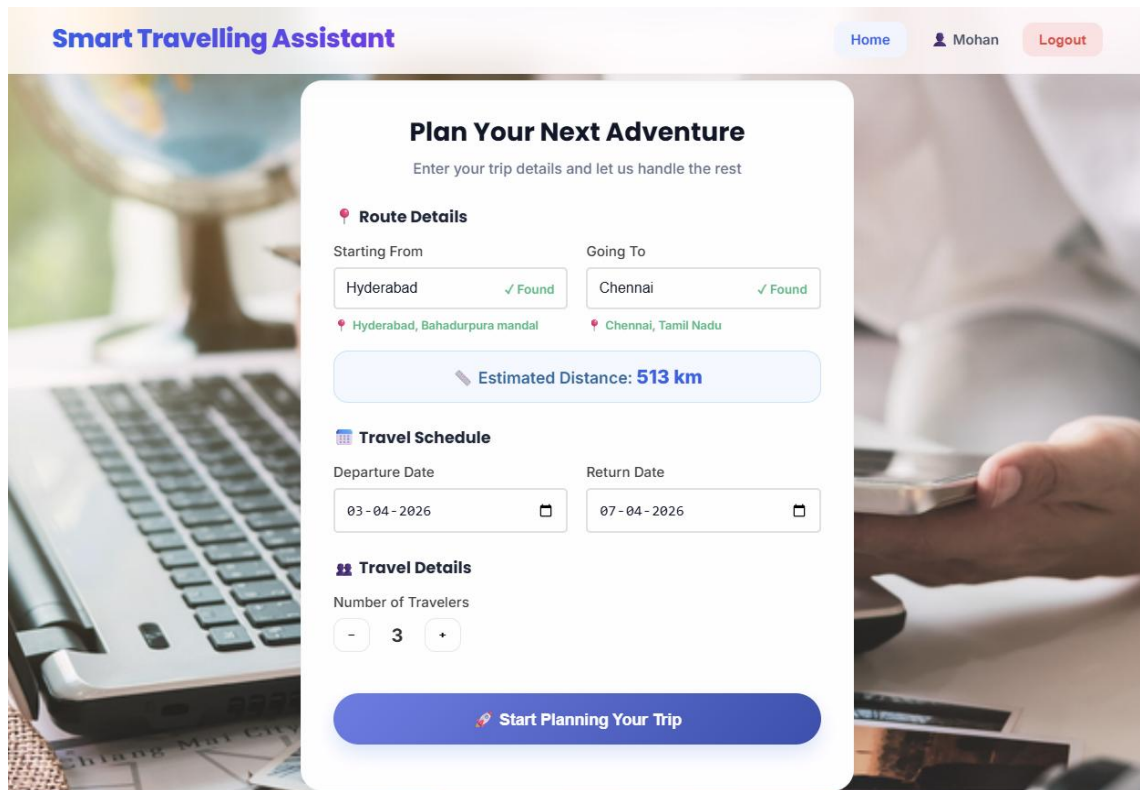
Password

Confirm Password

Register

Sign in as Mohan
mohangantu55@gmail.com

Figure 5.3 Register Page



Smart Travelling Assistant Home Mohan Logout

Plan Your Next Adventure

Enter your trip details and let us handle the rest

Route Details

Starting From: ✓ Found Hyderabad, Bahadurpura mandal

Going To: ✓ Found Chennai, Tamil Nadu

Estimated Distance: 513 km

Travel Schedule

Departure Date: 📅

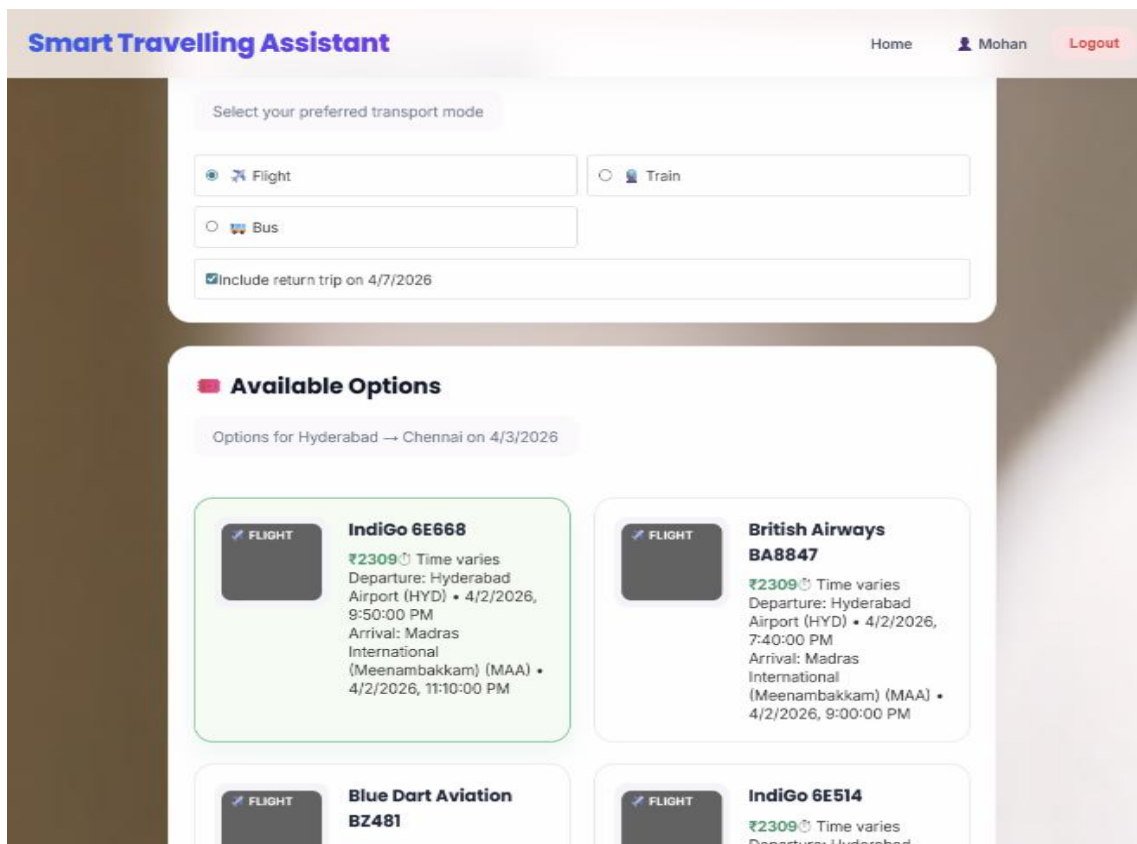
Return Date: 📅

Travel Details

Number of Travelers: - +

[Start Planning Your Trip](#)

Figure 5.4 Input Form Page



Smart Travelling Assistant Home Mohan Logout

Select your preferred transport mode

☒ Flight ☐ Train ☐ Bus

☒ Include return trip on 4/7/2026

Available Options

Options for Hyderabad → Chennai on 4/3/2026

Mode	Flight	Price	Time	Departure	Arrival
FLIGHT	IndiGo 6E668	₹2309	Time varies	Hyderabad Airport (HYD) • 4/2/2026, 9:50:00 PM	Madras International (Meenambakkam) (MAA) • 4/2/2026, 11:10:00 PM
FLIGHT	British Airways BA8847	₹2309	Time varies	Hyderabad Airport (HYD) • 4/2/2026, 7:40:00 PM	Madras International (Meenambakkam) (MAA) • 4/2/2026, 9:00:00 PM
FLIGHT	Blue Dart Aviation BZ481	₹2309	Time varies	Hyderabad Airport (HYD) • 4/2/2026, 9:50:00 PM	Madras International (Meenambakkam) (MAA) • 4/2/2026, 11:10:00 PM
FLIGHT	IndiGo 6E514	₹2309	Time varies	Hyderabad Airport (HYD) • 4/2/2026, 9:50:00 PM	Madras International (Meenambakkam) (MAA) • 4/2/2026, 11:10:00 PM

Figure 5.5 Transportation Page Part 1

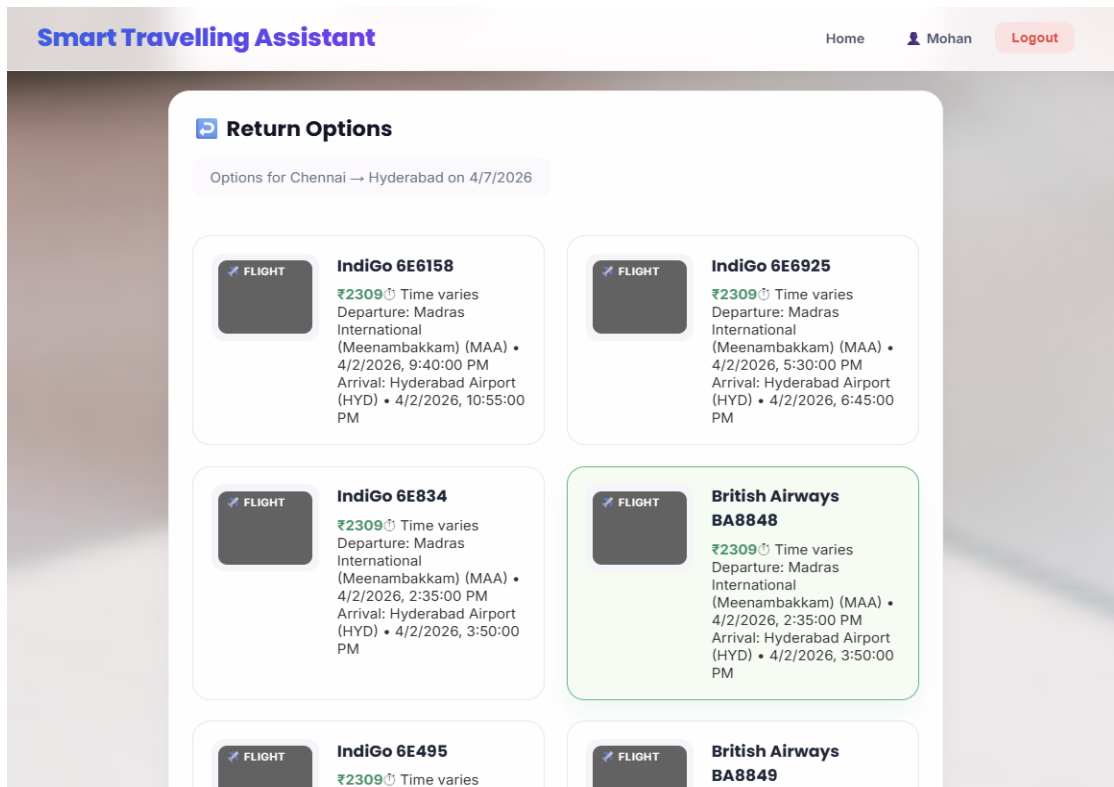


Figure 5.6 Transportation Page Part 2

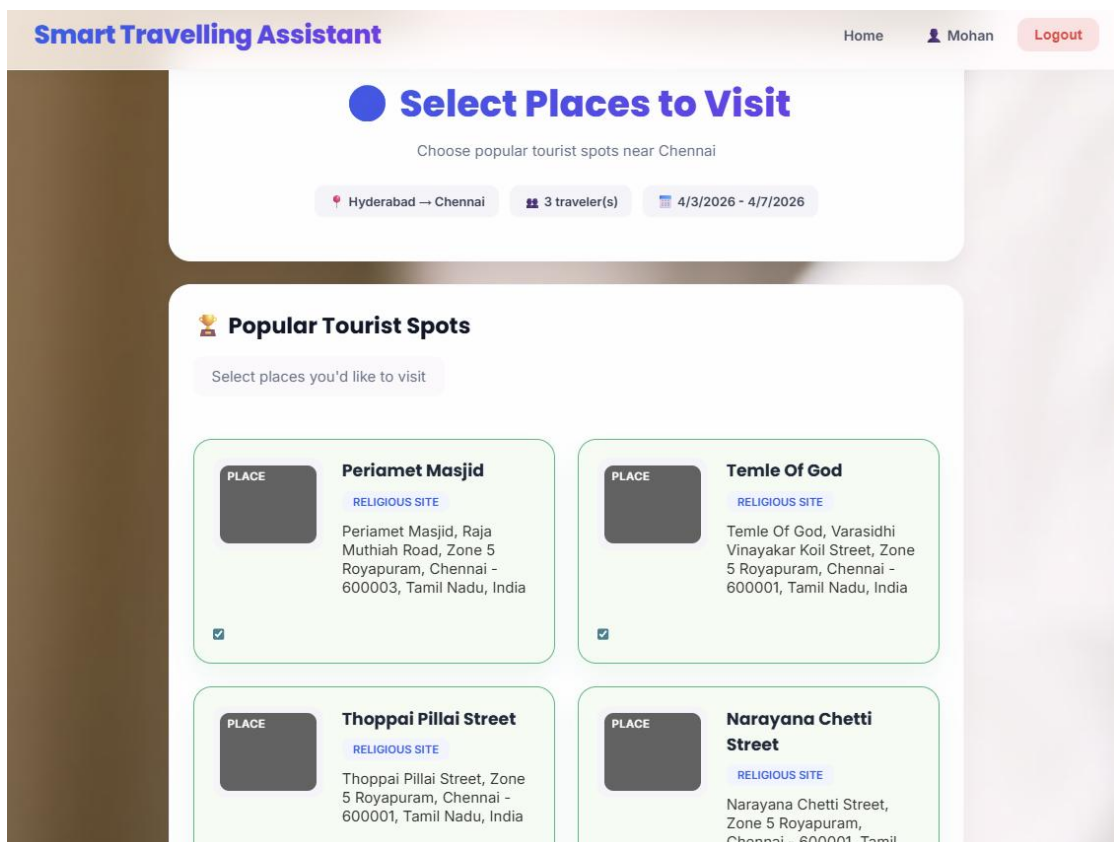


Figure 5.7 Places Page Part 1

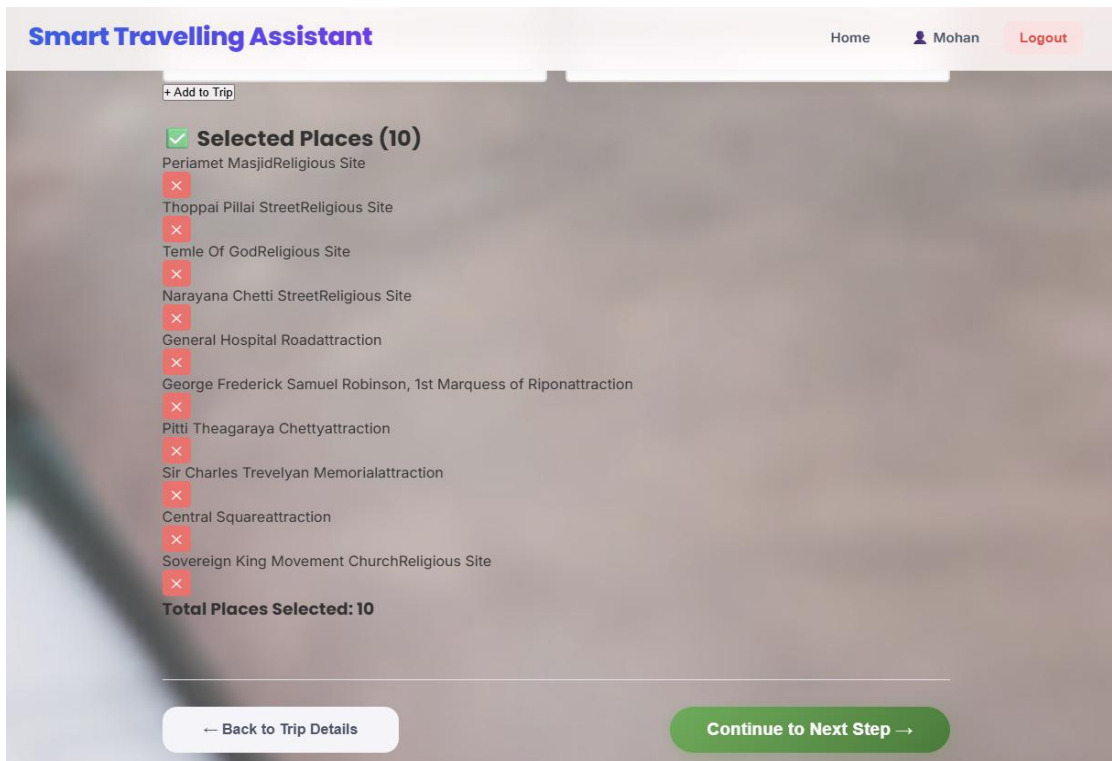


Figure 5.8 Places Page Part 2

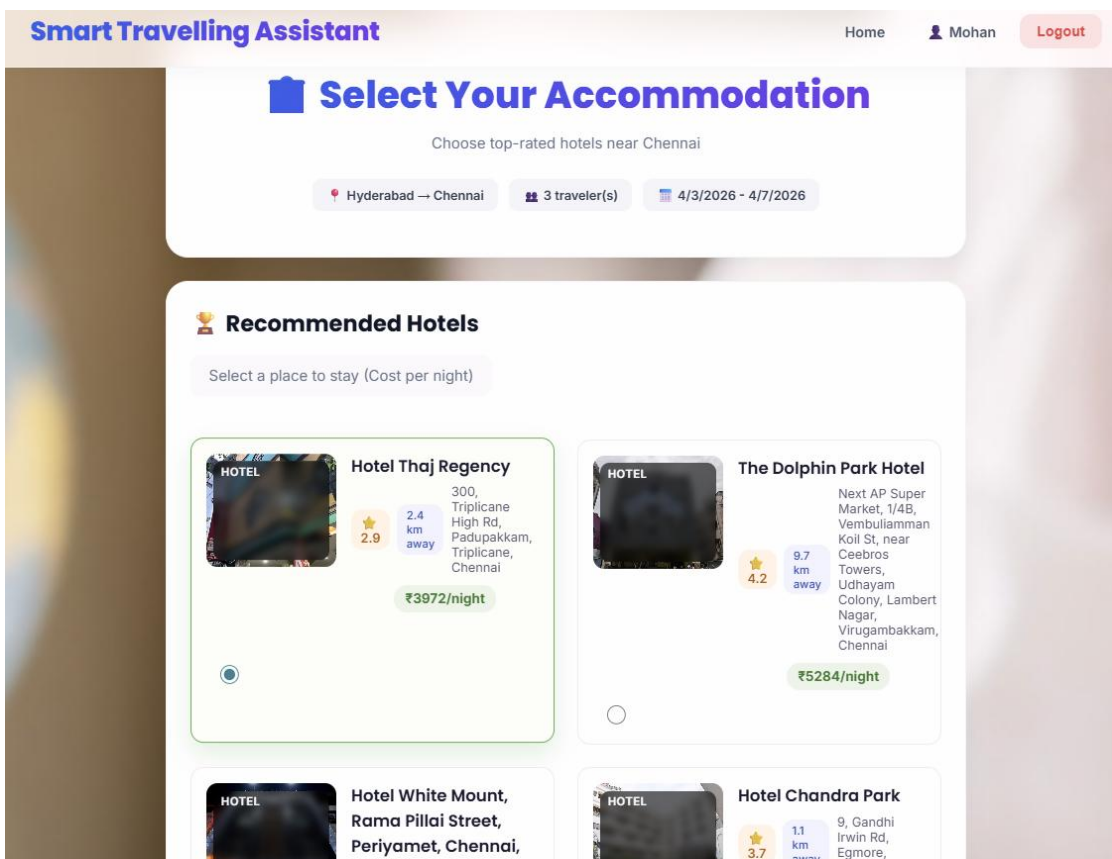


Figure 5.9 Accomodation Page

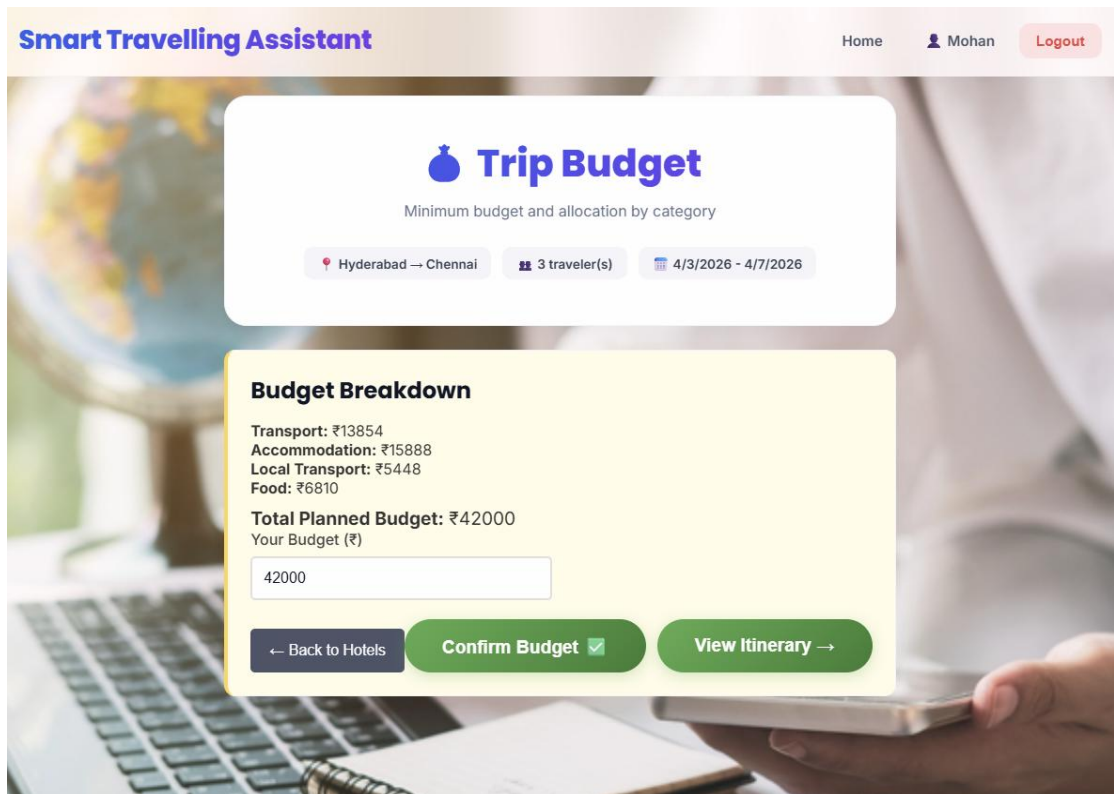


Figure 5.10 Budget Page

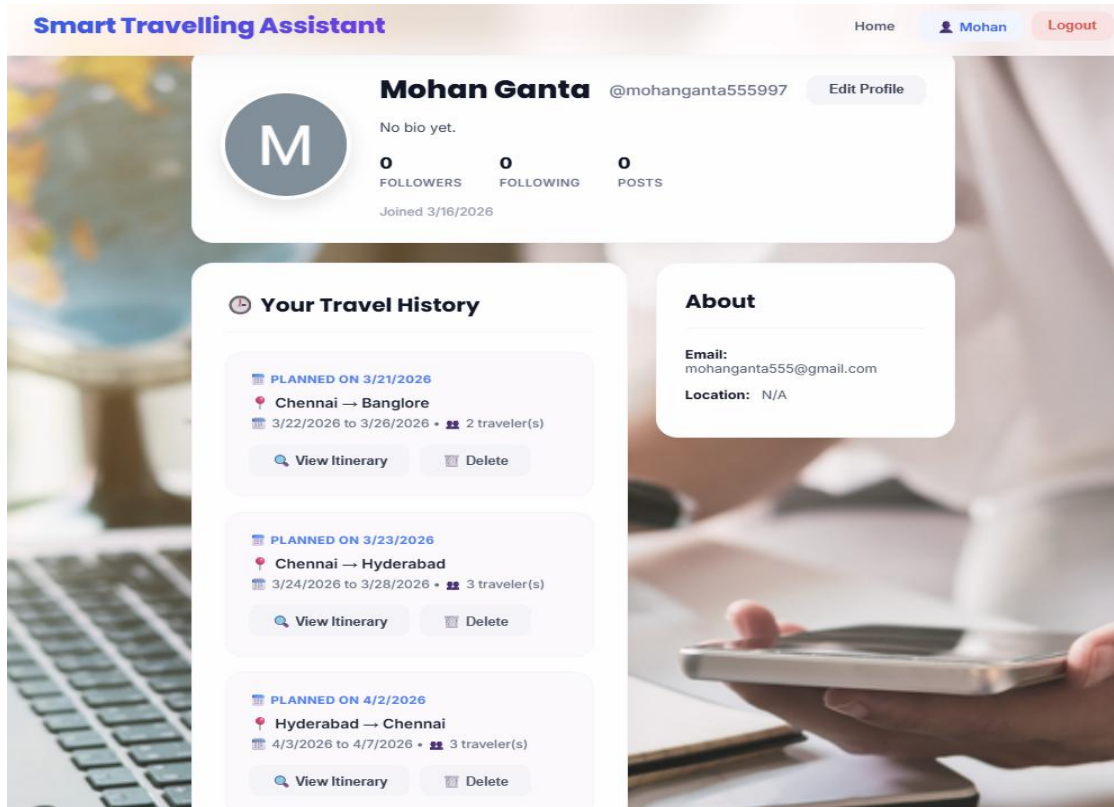


Figure 5.11 Profile Page

5.1.3 Result Analysis

The results obtained from the implementation of the Smart Travelling Assistant clearly demonstrate that the system successfully fulfills its intended objectives and provides an efficient solution for modern travel planning. The application effectively processes user inputs such as destination, budget, travel duration, and number of travelers to generate personalized and structured travel itineraries.

One of the major strengths of the system is its ability to integrate multiple real-time APIs, which ensures that the information provided to users—such as flight details, hotel recommendations, and route data—is accurate, up-to-date, and reliable. This significantly enhances the quality of the generated travel plans compared to static or manually created itineraries.

The implementation of the Time-Proximity Clustering Logic plays a crucial role in improving the practicality of the system. By organizing tourist attractions based on geographical proximity and estimated travel time, the system generates logically optimized and feasible day-wise schedules.

This minimizes unnecessary travel time and ensures efficient utilization of each day, thereby improving the overall travel experience.

From a performance perspective, the system exhibits fast response times and smooth interaction between the frontend and backend components. The use of modern technologies such as React and

Node.js ensures efficient data handling and seamless communication, resulting in a responsive and dynamic user interface.

The application also excels in terms of usability and user experience. The interface is designed to be intuitive, simple, and easy to navigate, allowing users with minimal technical knowledge to interact with the system effectively. Features such as the user dashboard and travel history management further enhance user engagement and convenience.

When compared to traditional travel planning methods, which often involve manual research and multiple platforms, the Smart Travelling Assistant significantly reduces time, effort, and complexity. By integrating various services into a single platform, the system provides a centralized and streamlined solution, leading to improved efficiency and higher user satisfaction.

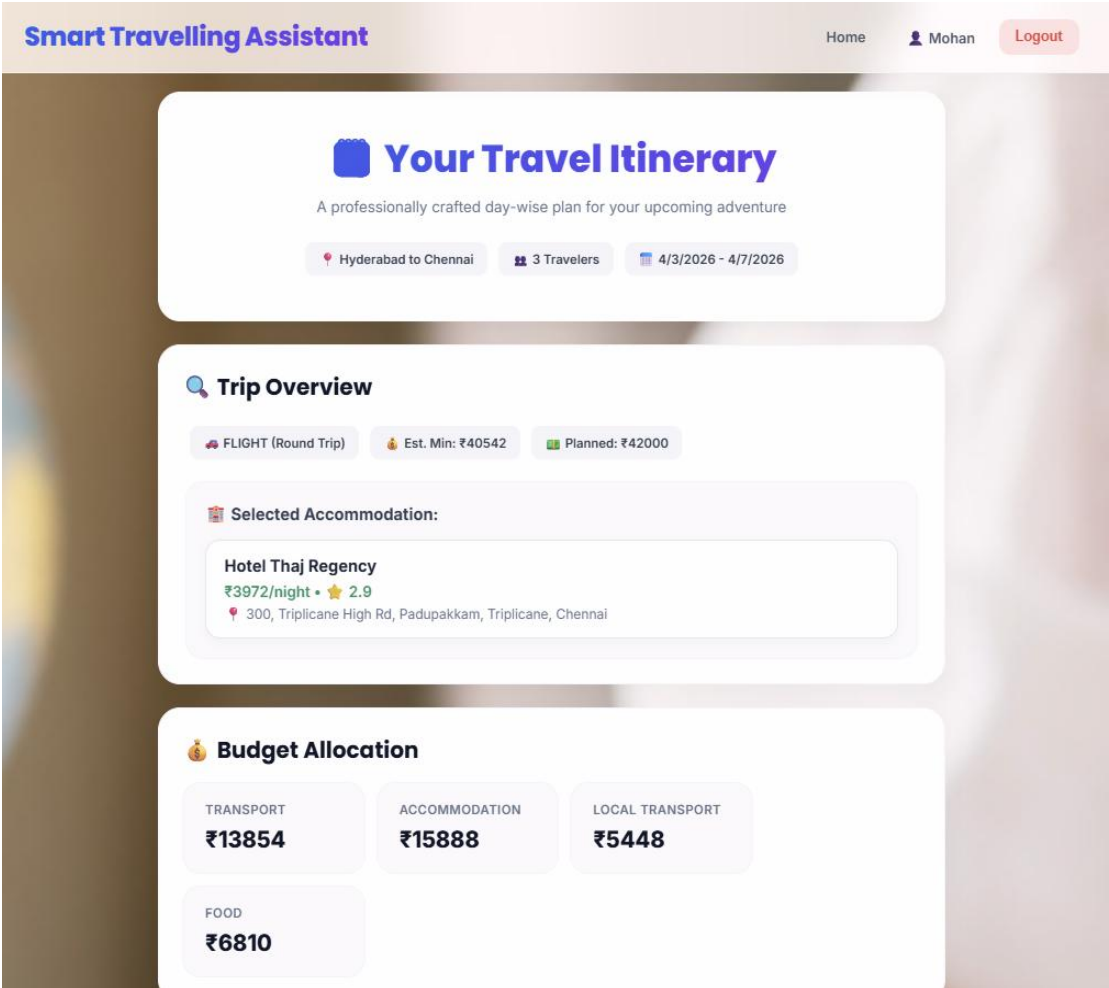


Figure 5.12 Itinerary Page Part 1

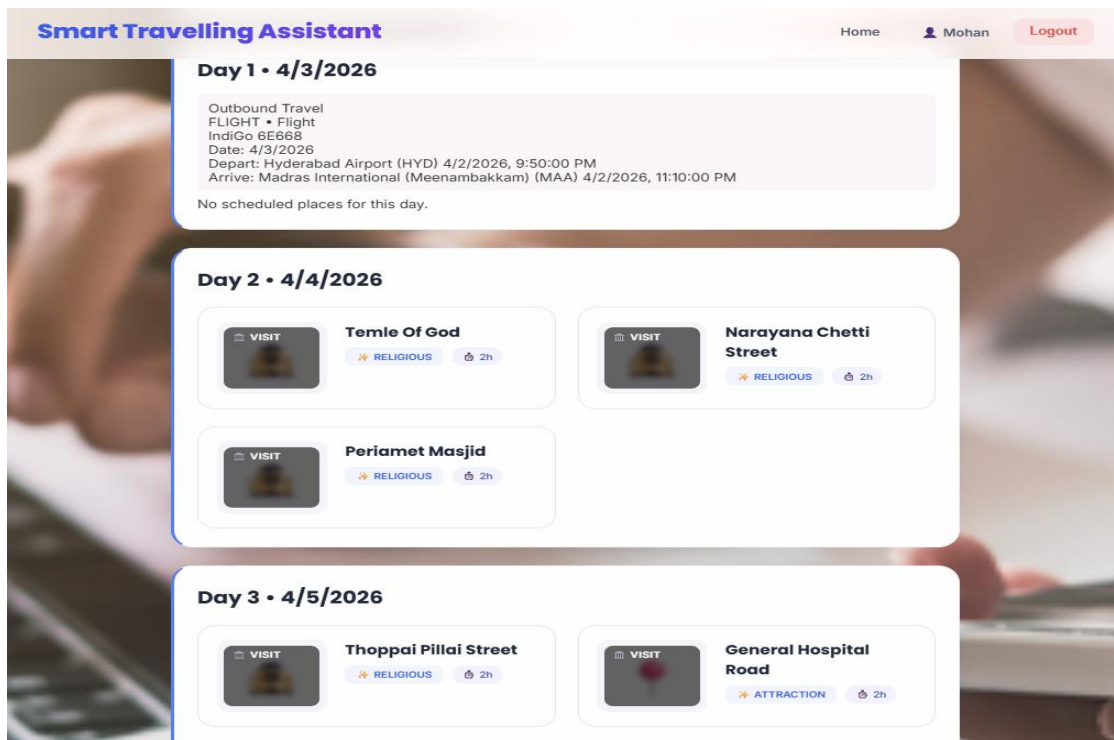


Figure 5.13 Itinerary Page Part 2

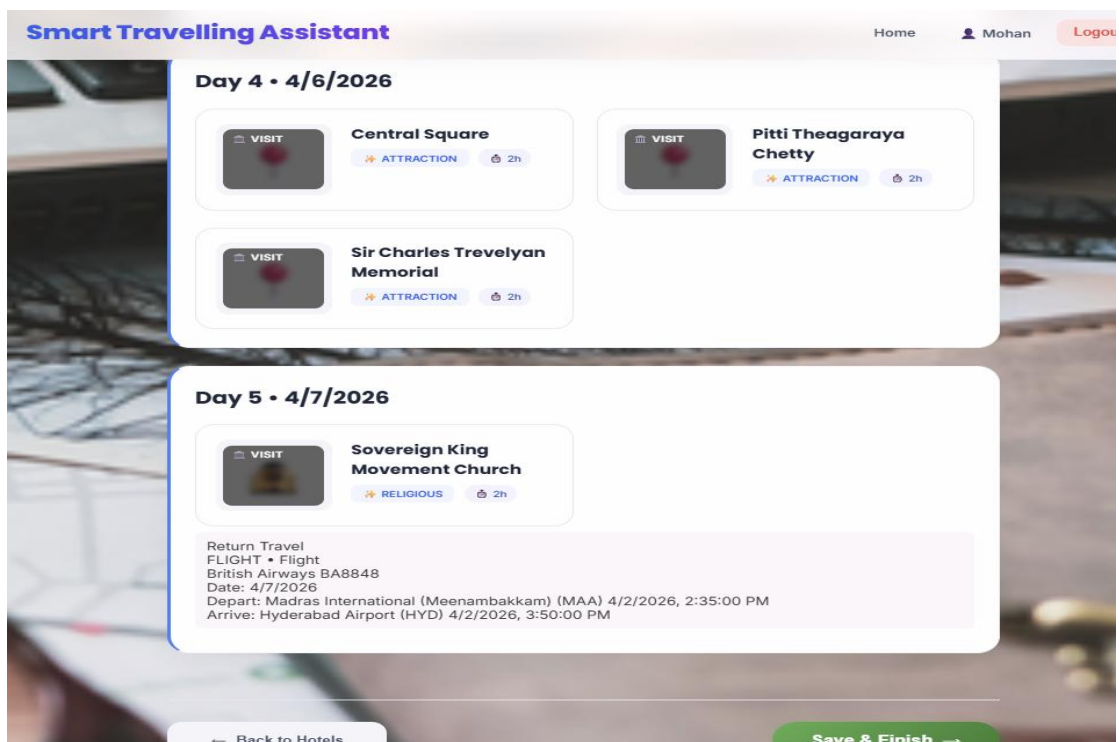


Figure 5.14 Itinerary Page Part 3

6. TESTING & VALIDATION

6.1 FEATURES TO BE TESTED

The following key features of the Smart Travelling Assistant are identified for testing to ensure proper functionality and performance:

- User Authentication System
 - Registration using email and password
 - Login using credentials and Google OAuth
 - OTP verification functionality
- Input Handling Module
 - Validation of user inputs such as destination, budget, and dates
 - Handling incorrect or incomplete inputs
- Itinerary Generation Module
 - Proper execution of Time-Proximity Clustering Logic
 - Accurate generation of day-wise travel plans
 - Logical distribution of activities
- API Integration
 - Retrieval of real-time data from AviationStack, Geoapify, OpenRouteService, and Google Places
 - Handling API failures or delays
- Dashboard Functionality
 - Display of user profile
 - Storage and retrieval of travel history
 - Deletion of saved plans
- Output Display

- Correct rendering of itinerary, hotels, and routes
- Responsive UI across devices

6.2 FEATURES NOT TO BE TESTED

The following aspects are not included in the testing scope due to limitations or external dependencies:

- Third-party API internal functionality (e.g., how Google Places processes data internally)
- External server reliability and uptime
- Payment systems (not implemented in this project)
- Advanced AI-based recommendations (future scope)
- Network-related issues beyond application control

These components are assumed to function correctly as they are managed by external providers.

6.3 TESTING TOOLS AND ENVIRONMENT

The system is tested using standard tools and environments to ensure reliability and performance.

Testing Tools

- Postman – for testing backend APIs
- Browser Developer Tools – for debugging frontend issues
- MongoDB Atlas Tools – for database validation
- Manual Testing – for user interface and usability

Testing Environment

- Operating Systems: Windows / Linux / macOS
- Browsers: Google Chrome, Mozilla Firefox, Microsoft Edge
- Hardware: Standard system with minimum 4GB RAM

6.4 DESIGN OF TEST CASES AND SCENARIOS

Table 6.1 Test Cases

Test Case ID	Test Scenario	Input Data	Expected Output	Result
TC01	User Registration	Valid email & password	User account created successfully	Pass
TC02	User Login	Correct credentials	Login successful	Pass
TC03	Invalid Login	Wrong password	Error message displayed	Pass
TC04	Input Validation	Empty destination field	Validation error shown	Pass
TC05	Itinerary Generation	Valid travel details	Day-wise itinerary generated	Pass
TC06	API Data Fetch	Request to external API	Data retrieved successfully	Pass
TC07	Dashboard Access	Logged-in user	User dashboard displayed	Pass
TC08	History Storage	Save travel plan	Plan stored in database	Pass
TC09	Delete History	Delete selected plan	Plan removed successfully	Pass

Test Scenarios

- Verify complete user flow from login to itinerary generation
- Test system behavior under invalid inputs
- Validate API response handling
- Ensure proper storage and retrieval of data
- Check UI responsiveness across devices

7. CONCLUSION & FUTURE ENHANCEMENT

7.1 PROJECT CONCLUSION

The Smart Travelling Assistant has been successfully designed and developed as an intelligent web-based solution aimed at simplifying and enhancing the travel planning process. The system effectively addresses the limitations of traditional travel planning methods by providing a centralized platform that integrates multiple services such as itinerary generation, route optimization, and accommodation recommendations. By utilizing modern full-stack technologies including React, Node.js, Express, and MongoDB Atlas, the application ensures high performance, scalability, and efficient data management.

A key strength of the system lies in its ability to generate personalized and structured day-wise travel itineraries based on user-defined inputs such as destination, budget, and travel duration. The implementation of the Time-Proximity Clustering Logic plays a significant role in producing practical and realistic travel schedules by minimizing travel time and maximizing location coverage. Furthermore, the integration of real-time APIs enhances the accuracy and reliability of the information provided to users.

The system also incorporates essential features such as secure authentication (Google OAuth and OTP verification), user profile management, and travel history tracking, thereby improving overall usability and user engagement. The deployment of the application on cloud platforms ensures accessibility, flexibility, and scalability. Overall, the project demonstrates a robust, efficient, and user-centric solution that meets the objectives of modern travel planning and provides a strong foundation for future advancements.

7.2 FUTURE ENHANCEMENT

Despite its current capabilities, the Smart Travelling Assistant offers significant opportunities for further enhancement and expansion. One of the major future improvements includes the integration of Artificial Intelligence and Machine Learning techniques to provide more intelligent and predictive travel recommendations based on

user behavior, preferences, and past travel history. This would enable the system to deliver a more personalized and adaptive user experience.

Another important enhancement is the incorporation of online booking functionalities, allowing users to directly book flights, hotels, and transportation services within the platform. This would transform the application into a complete end-to-end travel management system. Additionally, the development of a mobile application (Android/iOS) would increase accessibility and provide users with on-the-go travel planning capabilities.

Further improvements may include the addition of real-time weather updates, budget tracking and expense management, multilingual support, and social sharing features. The system can also be enhanced by integrating advanced analytics to provide insights into travel trends and user preferences. Moreover, improving system performance through caching mechanisms and optimizing API usage can further enhance efficiency.

In the long term, the Smart Travelling Assistant can evolve into a comprehensive intelligent travel ecosystem by incorporating emerging technologies and expanding its feature set, thereby providing a more seamless, efficient, and enriched travel planning experience.

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